

Brian Wonjun Lee

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EDUCATION

University of Pennsylvania

BA in Computer Science & Design (Double Major), Benjamin Franklin Scholars

Philadelphia, PA

Expected May 2027

Relevant Coursework: Interactive Computer Graphics, Computer Animation, Introduction to Algorithms, 3-D Computer Modeling

EXPERIENCE

Software Engineer Intern

Aleph Lab (Y Combinator)

June 2026 – Present

San Francisco, CA (Remote)

- Built a push-notification retention engine at a 4-engineer YC startup (Supabase triggers, Deno edge functions, FCM), instrumenting a treatment/holdout A/B framework to measure return-rate lift
- Designed a typed SDK putting a production game-AI agent behind 4 versioned contract seams (zod, JSON Schema, 128+ tests)
- Merged 15+ PRs in 3 weeks (React Native, Next.js, Deno), including an auth fix closing 4 unauthenticated production edge functions

Software Engineer Intern

BitMango (Puzzle1 Studio)

June 2025 – Aug 2025

Pangyo, South Korea

- Resolved 100+ QA-reported bugs across a live portfolio of 50+ Unity titles, shipping prefab and C# fixes in production
- Upgraded SDKs and platform modules fleet-wide to meet evolving app-store requirements and sustain production availability

Project Lead & Product Designer

Penn Spark (Red Team)

Jan 2025 – Present

Philadelphia, PA

- Led a 15+-person cross-functional team of designers and developers to ship client-facing 0-to-1 products
- Shipped polished, user-centered features across semester-long client project cycles

Military Interpreter

Capital Defense Command (ROK Army)

Dec 2022 – June 2024

Seoul, South Korea

- Provided real-time English-Korean interpretation and translated sensitive briefings in high-security, multi-agency environments

Database Engineer Intern

IT-Farm

June 2022 – Aug 2022

Seongnam, South Korea

- Designed and implemented a SQL database prototype managing 1M+ data entities using Oracle and SQLite
- Processed and organized semiconductor datasets from 20+ client companies, including Samsung, SK, and LG

PROJECTS

[Mini Minecraft](#) | C++, OpenGL, GLSL

Apr 2026

- Engineered a voxel engine from scratch: 7 procedural biomes from layered Perlin/FBM/Voronoi noise with smoothstep blending
- Implemented 3D Perlin cave systems with trilinear interpolation, fluid physics, and post-process tint overlays
- Built the render pipeline: 7x7 PCF soft shadows, screen-space reflections, vertex AO, and a dynamic day-night cycle

[Passenger](#) | Unreal Engine 5, HLSL, Maya

Mar 2026 – Apr 2026

- Wrote 2 HLSL post-process shaders (anisotropic Kuwahara, cell shader) sampling GBuffer channels for stylized rendering
- Rigged and animated a character in Maya (IK limbs, spline-IK spine) driving an Unreal Animation Blueprint
- Rendered with 64 TSR samples per frame at 2560x1080 via Movie Render Pipeline, hardware ray tracing over Lumen GI

[Mini Maya](#) | C++, OpenGL, Qt

Feb 2026 – Mar 2026

- Implemented a half-edge mesh editor in C++: edge split, extrusion, and Catmull–Clark subdivision on an OpenGL pipeline

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Java, C++, C#, SQL, OCaml

Frameworks: React, Next.js, Node.js, React Three Fiber, Tailwind CSS, Astro

Tools: Unreal Engine 5, Unity, OpenGL, GLSL, HLSL, Maya, Blender, Substance Painter, Figma, Git